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PROBLEMS OF PLASTIC GLIDING

By E. OROWAN, Birmingham

ABSTRACT: A discussion is given of the rate of flow in metal single crystals. Flow is believed to be due to the presence of dislocations; the rate of production and rate of movement of dislocations are treated.

§ 1. THE KINEMATICS AND DYNAMICS OF GLIDING

A FEW years ago the kinematics (crystallography) of plastic gliding was a self-contained domain that could be treated without paying much attention to dynamical questions. Recent researches, in particular those made by Prof. Andrade and his collaborators⁽¹⁾, have shown, however, that the operative glide plane and glide direction vary, in general, with the state of stress and with the temperature. Thus further progress in the crystallography of gliding has become largely dependent upon the development of our knowledge of its dynamics.

§ 2. THE SIGNIFICANCE OF THE STRESS-STRAIN CURVE

The development of the dynamics of gliding was, for many years, heavily impeded by a seemingly plausible, but actually wrong, hypothesis inherited from engineering mechanics. This hypothesis assumed that the plastic properties of materials can be described by stress-strain curves; in other words, that the dynamic law of gliding consists of a relationship between stress and strain. The empirical basis of this assumption (which we shall call the "static" hypothesis) was the following observation. If we apply a constant load to a commercial metal, the extension as a function of time will be given by a curve like figure 1*a*. There is, first, a period of rapid extension, lasting e.g. for a few seconds; after this the specimen has settled down to a comparatively constant length, and a period of very slow extension follows which, in general, lasts as long as the load is applied. Now the extension reached after the first period is plotted against the load in the stress-strain curve; the slow extension of the second period, the "creep", is made the subject of a special study independent of that of the stress-strain curve.

We see that the use of the stress-strain curve is an approximation permissible if the duration of loading is short enough to prevent appreciable creep, but longer than the first period of rapid extension. Thus stress-strain curves cannot be used for processes like rolling where the deformation takes place in a less time than the first period of extension.

When the study of the plasticity of metal single crystals began, the use of stress-strain curves was adopted without due realization of their roughly approximative nature, and for many years it was not noticed that this procedure does not even represent a rough approximation to the actual behaviour of single crystals. If we

apply a constant load to a single crystal, we obtain in many cases extension-time curves like figure 1*b*. This curve shows no sharp bend; it is impossible to determine without arbitrariness when the rapid extension ceases and when creep begins, and the crystal does not settle down to a more or less defined length. How was it, then, possible to obtain stress-strain curves at all? This was done in the following way. The extensometers used were so designed that the load increased continuously with time, and simultaneous readings of load and extension were used for plotting a stress-strain curve, without regard to whether the crystal would attain, at least approximately, a constant length under a constant load. It was not realized that any highly viscous Newtonian liquid, for example soft pitch or treacle, would give in this way stress-strain curves that are exactly like the typical stress-strain curves of metals, although there is no strain hardening, and no physical relationship whatever between stress and strain.

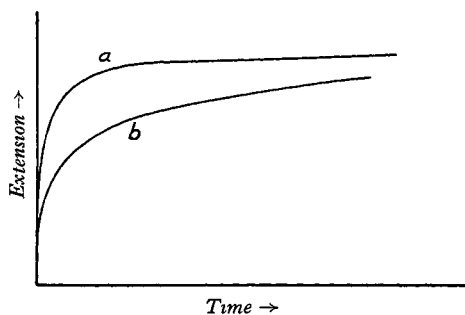


Figure 1.

It was well known, of course, that crystals “flow” under a constant load; this, however, was interpreted in the following way. It was assumed that, for any strain, the crystal would possess a certain “strength”, represented by the corresponding ordinate of the stress-strain curve, and further deformation could take place only if the stress exceeded this strength. Plastic deformation produces internal distortions which make further deformation more difficult and thus increase the strength (“strain hardening”). The internal distortions produced by gliding are removed by thermal agitation: a strain-hardened crystal becomes soft again if it is kept for a sufficient time at a sufficiently high temperature. Now this “thermal softening” goes on during the extension test as well; consequently, if a constant load is applied to the crystal, plastic deformation will go on at such a rate that the thermal softening per unit time will just be compensated by the strain hardening produced by the deformation that takes place per unit time. Thus, according to the static hypothesis, any flow would be caused by thermal softening and no true flow would exist, that is, no flow independent of strain hardening and thermal softening.

However, it is easy to show that most observed cases of plastic flow certainly represent a true flow that cannot be explained by thermal softening^(13, 15). The time rate of softening decreases very rapidly with decreasing temperature. As an atomic

rearrangement process, activated by thermal agitation, softening shows the same typical temperature dependence as, for example, the velocity of chemical reactions, decreasing to about half or a third of its value if the temperature decreases by 10° . Now in the case of tungsten crystals the time constant of thermal softening reaches the value of one day probably at some temperature between 500° and 1000° C.; to be on the safe side, we assume that it is one day at 500° and that it is halved if the temperature decreases by 10° . The time constant at room temperature is then about a million million years; in other words, there is no thermal softening whatever at room temperature, and the whole amount of the abundant flow which tungsten crystals show at room temperature⁽²⁾ is a true flow.

Another striking case is the flow of cadmium crystals at 1.2° K.⁽¹⁰⁾

The existence of a true flow shows that plastic gliding obeys a "dynamic" law; that is to say, the time rate of strain, not the strain itself, is a function of stress. The time rate of strain will depend, besides, on temperature and on the "state of strain hardening" of the crystal. The latter is not simply a function of the strain, but depends on all values which the strain and the temperature (and probably, the rate of strain) have had in the course of time.

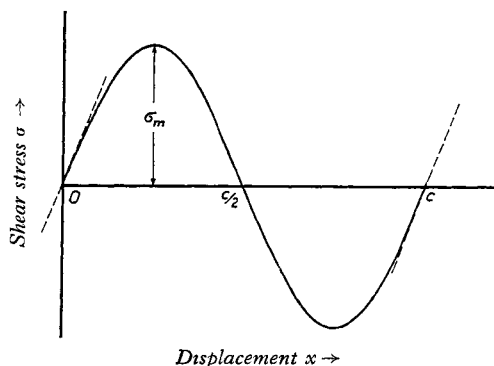


Figure 2.

§ 3. THE EXPERIMENTAL PROGRAMME

It would seem at first that the question whether true flow exists or not is rather an academic one. Actually, however, the answer to this question predetermines not only the programme of the measurements to be made but also the design of the extensometers to be used.

If there is no true flow, then

- (a) the fundamental measurement to be made is that of the stress-strain curve; during this measurement
- (b) thermal softening must be kept as small as possible; and
- (c) the extension test must be continued until rupture occurs, in order to obtain the whole stress-strain curve.

(d) In the absence of thermal softening the rate of loading and its fluctuations are irrelevant; therefore any type of extensometer may be used.

(e) After finding the stress-strain curve, the effect of thermal softening is studied.

If true flow exists, on the other hand,

(a) the fundamental measurement is that of the rate of flow at different stresses and temperatures; during this measurement

(b) strain hardening must be kept as small as possible, and therefore

(c) the measurement must be made with extensions as small as possible.

(d) Extensometers must be used which allow the load to be kept constant and uninfluenced by the extension of the specimen and by inertia effects.

(e) After finding the rate of flow as a function of stress and temperature for the unhardened crystal, the effect of strain hardening can be studied.

This summary explains why, unfortunately, the majority of the dynamic measurements made under the influence of the static hypothesis (that is, the majority of the measurements made up to the present) are of little or no use; in fact, the main result from that period is that an explanation of plastic gliding on that basis is impossible, and an entirely new theoretical and experimental departure must be made.

§ 4. THE EXTENSOMETER

The first task is to design an extensometer suitable for the experimental programme set by the dynamic conception of gliding. The extensometers used so far belong partly to the "screw-and-spring" type in which one end of the specimen is pulled by a screw and the other connected with a spring which produces and at the same time measures the load, and partly to the weight-loading or to the electro-magnetic type. In the first type, the spring contracts when the specimen extends, and therefore the load cannot be kept constant. In the second type, heavy masses are connected with the moving end of the specimen; owing to the jerky nature of extension in many important cases, the inertia of the masses produces large fluctuations of the load.

In the Physics Department of the University of Birmingham an extensometer has been built to suit the requirements of the new experimental programme. This is a weight-loading instrument in which between the weight and the specimen a soft spring is inserted; thus the load is independent of the extension, and at the same time inertia effects are practically eliminated by the spring. The extension can be interrupted after it has reached a previously adjustable amount, and it is photographically recorded with a low and a high magnification simultaneously.

It will be, of course, some time before a sufficient number of experimental results are available to allow us to draw a coherent picture of the plastic behaviour of crystals. In the meantime it is appropriate to put the question: What safe conclusions about the mechanism of plastic gliding can be drawn from the experience available at present? In what follows an attempt will be made to answer this question.

§ 5. THE DISCREPANCY BETWEEN THEORETICAL AND PRACTICAL STRENGTHS: DISLOCATIONS

The first fundamental problem of plastic gliding relates to the discrepancy between microscopic or "theoretical" and macroscopic strength^(5,17). There is a simple and safe way to obtain an estimate of the maximum shear stress which the lattice can withstand. We choose a lattice plane as a glide plane, and in it a row of atoms as the direction of gliding. Let d be the distance between two neighbouring lattice planes of this sort, and c the distance between two atoms in the glide direction. If a shear stress σ acts in the glide plane parallel to the glide direction, two neighbouring planes suffer a relative displacement x ; the shear strain is, then, x/d . For small values of x/d Hooke's law holds, and we have

$$\sigma = G \cdot x/d, \quad \dots\dots(1)$$

where G is the modulus of shear. In figure 2, where σ is plotted as a function of x , (1) is represented by the dotted straight line going through the origin.

If we assume that the distortion of the lattice remains strictly homogeneous, however large the strain, we see that for $x=c/2$ the stress σ must vanish, since all atoms of a plane are then symmetrically between two atoms of a neighbouring plane. For $x=c$ we have again the undistorted lattice, and the relationship between σ and x in the neighbourhood of $x=c$ is given by a straight line representing Hooke's law. Thus, obviously, σ must be approximately a sine function of x :

$$\sigma \simeq \sigma_m \cdot \sin \frac{2\pi}{d} x, \quad \dots\dots(2)$$

as shown in figure 2; σ_m is the maximum shear stress which the lattice can withstand; in other words, the microscopic or "theoretical" shear strength of the glide system considered.

For small values of x/c , (2) becomes approximately

$$\sigma \simeq \sigma_m \cdot \frac{2\pi}{d} x; \quad \dots\dots(2a)$$

this equation must be the same as (1), and comparing its constants with those in (1), we obtain

$$\sigma_m \simeq \frac{G}{2\pi} \cdot c/d. \quad \dots\dots(3)$$

c and d being, in general, nearly equal, we have

$$\sigma_m \simeq G/2\pi, \quad \dots\dots(3a)$$

that is to say, the theoretical shear strength is approximately the modulus of shear, divided by 2π . This quantity is, for single crystals of the common metals, about 100 to 10,000 times greater than the actually observed macroscopic yield stress. What is the cause of this discrepancy?

First of all, there is discrepancy of the same magnitude between the microscopic and macroscopic tensile strength of brittle solids, like glass. This discrepancy has been explained by A. A. Griffith in a very satisfactory way which has been verified experimentally; it is caused by minute cracks or other defects, in the neighbourhood of which the local stress is much higher than the average stress in

the material. However, this explanation cannot be applied directly to plastic gliding. In the case of tensile strength, the stress concentration travels with the edge of the crack across the specimen when the crack begins to spread; if, however, the increased local shear stress at the edge of a crack starts gliding there, the edge of the crack remains where it was and does not cause a travelling stress concentration that would explain how the glide process is able to spread across the specimen.

It is easy to see, however, that a glide process, once started, itself creates the inhomogeneity necessary for its propagation. Suppose that a glide process started at P in figure 3 (figure 3a shows the glide plane, 3b its side view) and that, at some

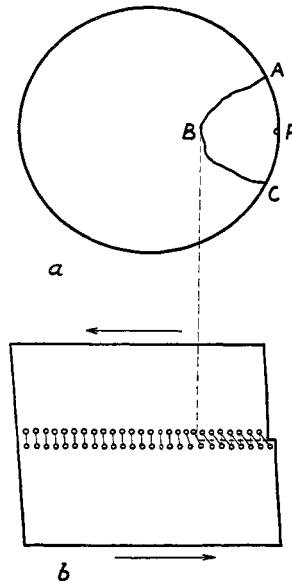


Figure 3.

moment, it has spread to the line ABC ; in consequence, at this moment ABC separates those parts of the glide plane which have already undergone gliding from those which have not. The atoms of the lattice planes which glide upon each other are *still* in phase outside ABC and *again* in phase inside ABC . Along ABC , however, there is a zone in which the atoms of the lattice planes on both sides of the active glide plane are out of phase; they are more or less lifted out of the potential troughs of the field of the opposite lattice plane. Thus a relatively small stress will be sufficient to throw the atoms in this zone into their next equilibrium position and so to move the zone, that is, to propagate the glide process over the glide plane.

The zone of disorder along the line ABC has been termed by Prof. G. I. Taylor a *dislocation*. We shall call the line ABC (which can be a closed line if gliding has started from a point inside the crystal) a *dislocation line*, and the area inside this line the *dislocated area*.

It is important to realize that the existence of dislocations is not a hypothesis invented for explaining the discrepancy between microscopic and macroscopic strength. That dislocations *must* exist follows directly from the fact that gliding cannot take place simultaneously over the whole glide plane but must begin at one point and spread from there. In this connexion it is of interest that dislocations in the present sense were first treated by Prandtl in 1913^(9, 20), long before the problem of macroscopic strength arose; he made use of them for a tentative theory of mechanical hysteresis. The above picture of the propagation of gliding over a glide plane was put forward by the author in his (unpublished) Thesis in 1929; the role of dislocations was independently recognized by others also, and the idea that gliding propagates over the glide plane by dislocations was introduced into the literature of plasticity in 1934 in simultaneous papers by Prof. Polanyi⁽¹⁸⁾, Prof. G. I. Taylor⁽²¹⁾ and the author⁽¹⁴⁾.

The presence of flaws and dislocations shows that the crystal cannot possess the theoretical strength calculated for a homogeneous lattice. Whether dislocations actually lead to the observed values of the yield point is, of course, a separate problem. The critical stress necessary for setting a dislocation line into movement can be calculated if the width of the dislocation zone is known; this difficult mathematical problem has only very recently been solved by Prof. Peierls (cf. his paper later in this volume).

§ 6. THE EVIDENCE FOR DISLOCATIONS: THERMAL HARDENING

The conception that a perfect lattice has an extremely high strength and hardness and that the relative softness of real crystals is due to imperfections has often met with the following objection. Since plastic deformation creates imperfections and annealing removes them, it might be expected that crystals would become softer by plastic deformation and harder by annealing. According to common experience, however, the opposite is the case: deformation creates strain hardening which can be removed by annealing.

It is not difficult to recognize the cause of this seeming contradiction. The imperfections which cause the softness of the crystal are of the type of cracks or dislocations. The free energy of these imperfections is comparatively low, and a very thorough annealing is necessary to influence them. Furthermore, the softness of the crystal will not show any considerable change until the very last of these defects is removed or its efficiency diminished. On the other hand, the imperfections which cause strain hardening consist, according to x-ray evidence, of drastic distortions and rotations of the lattice; they are easily removed by annealing and to the extent that they are removed it is found that strain hardening diminishes. Thus, in general, only the hardening type of imperfections will be influenced by the usual kind of annealing, and, on the other hand, plastic deformation will not cause softening because, with most materials, softening imperfections are always present.

However plausible this explanation, it is very desirable that experimental evidence should be produced showing that the hardness of crystals can actually

be increased by removing imperfections, just as the tensile strength of glass⁽⁷⁾, rock salt⁽⁸⁾, and mica⁽¹¹⁾ could be considerably increased by eliminating surface defects. This experimental evidence was produced for zinc crystals⁽¹⁴⁾. A small extension in a spring-and-screw type extensometer, slightly exceeding the yield point, gave a stress-strain curve like curve 1 in figure 4; the load was then removed and the crystal given a rest of 24–48 hr. at room temperature (at which zinc crystals show comparatively rapid thermal softening). After this the crystals showed no trace of plastic deformation up to stresses far above the highest stress applied in the previous experiment—in one case, for example, up to a stress 30 per cent above

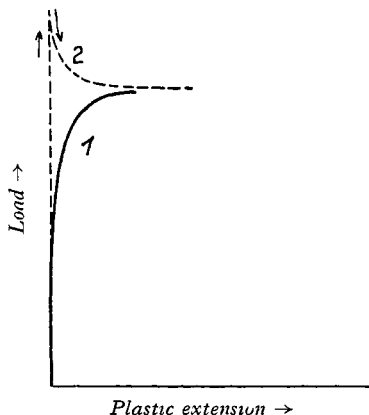


Figure 4.

the previously ascertained yield limit of the crystal. Frequently the crystal remained for many minutes under such high loads without any trace of extension; then a sudden dilatation occurred, the spring of the extensometer contracted, and the load fell until the crystal flowed slowly at the previously observed yield stress. This process is indicated by curve 2 in figure 4. Subsequent repetition of the experiment gave the first curve again: no sharp elastic limit could be observed and the crystal extended freely at its original yield point.

This experiment proves that the hardness of a crystal can increase by annealing and that it can suddenly decrease by plastic deformation. Consequently, the softness of real crystals can be attributed to imperfections and not to inherent structural characteristics of the lattice.

§ 7. THE EVIDENCE: STRAIN SOFTENING

An effect very similar to the “strain softening” and “thermal hardening” of zinc crystals is the well-known sudden yielding of soft steel; with the usual extensometers the stress drops after reaching the yield point (figure 5). It is still a subject of discussion among engineers how far this stress-strain curve is influenced by the design of the extensometer^(19, 22); as we have seen, the usual extensometers, as well as the usual methods of investigating the plasticity of metals, are essentially unsuitable for obtaining results with a physical meaning, since they are founded upon

the conception that no true flow exists. The practical justification of these methods is the intense strain hardening shown by most commercial metals; owing to this circumstance, the stress-strain curve obtained with a rapid extension is nearly independent of the rate of loading. This is not the case, however, for the initial stage of deformation of soft steel where the usual methods and conceptions lead to manifest contradictions.

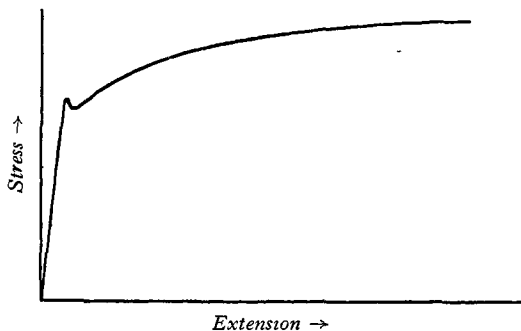


Figure 5.

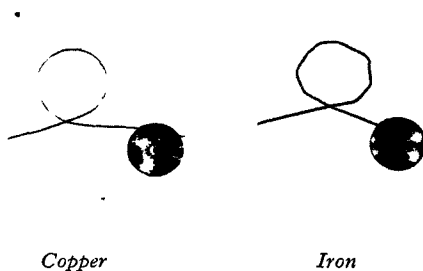


Figure 6.

We can easily prove, however, that plastic deformation actually makes annealed soft steel softer. If we bend a wire around a finger, the shape of the ring obtained will be different according to whether the wire hardens or softens by plastic deformation. If it hardens, bending will continue at those points where bending has not yet taken place and which, therefore, are still soft; the ring will be, then, of continuous curvature. If the wire is softened by deformation, bending will continue at the points where it began and which, therefore, have become softer; in this case the ring will be a polygon. Figure 6 shows that copper hardens and soft iron softens by bending. After the experience with zinc crystals there is scarcely any doubt that this strange behaviour of soft steel is due to the fact that the imperfections of the dislocation type are sensibly affected by annealing.

§ 8. THE PROPAGATION OF DISLOCATIONS

According to the previous considerations, plastic gliding involves two consecutive processes: the start of the glide process and its propagation. If one of these processes is much more difficult than the other, it alone will determine the velocity of plastic flow.

Let us assume that the amount a of glide in the glide process starting from the point P in figure 3 is one atomic distance in the glide direction.* After it has spread over the whole glide plane, the relative displacement of the two parts of the crystal separated by the active glide plane is a ; in an intermediate stage the displacement is aZ/A , where Z is the dislocated area and A the area of the glide plane.

The relative velocity of the two parts is $\frac{a}{A} \frac{dZ}{dt}$. The number of dislocation lines per unit length of the crystal, measured in a direction perpendicular to the glide plane, is NA , if N is their number per unit volume. Thus the rate of flow is

$$dx/dt = NA \frac{a}{A} \overline{dZ/dt} = Na \overline{dZ/dt}, \quad \dots(4)$$

where $\overline{dZ/dt}$ is the mean value of dZ/dt for all dislocation lines in the crystal.

In the special case when N is constant we have for the time average of the rate of flow

$$\overline{dx/dt} = Na \frac{1}{\tau} \int_0^\tau \overline{dZ/dt} dt = Na \frac{1}{\tau} A, \quad \dots(5)$$

if τ is the time in which a dislocation line spreads over the whole glide plane. Denoting by n the number of glide processes starting per unit time per unit length of the crystal (measured perpendicularly to the glide planes), we have

$$NA = n\tau, \quad \dots(6)$$

and from (5) we obtain $\overline{dx/dt} = na$(7)

Equation (7) can, of course, only be applied if the time τ of propagation, over which (5) is averaged, is much less than the duration of the extension test; only in this case is the rate of flow independent of the rate of movement of the dislocations and dependent only on the rate at which they are produced. The first attempts at a theory of plastic gliding actually assumed this to be the case; in other words, they assumed that the creation of dislocations is a process much slower than their propagation and, therefore, that it alone determines the rate of plastic flow. Subsequent experiments showed, however, that this is very far from being the truth. In these experiments⁽³⁾ the jerky extension of zinc crystals was photographically recorded with high magnification and the start as well as the further development of the glide process was made directly perceptible. The recorded curves (such as figure 7) reveal two important things. First, that the glide process starts suddenly and gradually falls off; the average duration of vigorous flow is at room temperature

* If it amounts to two or more atomic distances, it can be considered as a superposition of two or more dislocations by one atomic distance each.

of the order of seconds; at 70° or 80° C. of fractions of a second; and at -10° or -20° C. of minutes. Secondly, that a single glide process, started obviously from one point of the crystal, produces gliding by thousands or even millions of atomic distances; in other words, a first impulse gives rise to a whole avalanche of dislocations. These are evidently not confined to a single glide plane, since the step-like glide markings upon the surface of the crystal are not sharp but rounded off^(14, 6); thus gliding in one glide plane can "infect" neighbouring glide planes, producing dislocations in them.

According to these experiments the conditions of equation (7) are not even approximately fulfilled, and the rate of flow must be calculated from the general

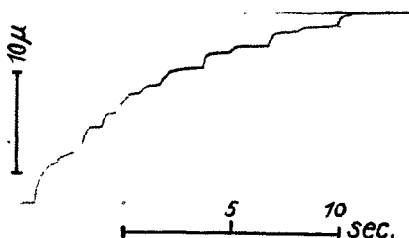


Figure 7.

equation (4). This represents a grave difficulty, since the rate of flow is given in (4) as a product of the number of dislocations per unit volume (or, roughly speaking, the number of active glide planes) and of the mean rate of increase $\overline{dZ}/dt$ of the dislocated areas, neither of which factors is directly observable. However serious this difficulty is, there is a hope that it can be surmounted by suitable experimental methods. If, in the course of an extension test, the stress is quickly altered by a small amount, the number of active glide planes will remain practically unaltered in many cases; whether this is so, can be checked by returning to the first stress again and seeing whether the rate of flow has its initial value. In this way the rate of propagation of the dislocations can be measured as a function of stress (and of temperature), and if this function is known, variations of N can be detected by using equation (4).

§ 9. THE ORIGIN OF DISLOCATIONS

The jerky extension of zinc crystals (which has also been observed with cadmium, rock salt, and other crystals) shows that gliding is more complicated than we might have assumed from consideration of the propagation of dislocations: the first glide step creates not one dislocation, but a large number of dislocations in different glide planes. This complication, however, is more than compensated by the opportunity given in the jerky extension of immediately perceiving the start of glide processes and thus being able to study the dependence of their frequency upon stress and temperature. This study could not previously be made, owing to the lack

of suitable extensometers; but it was observed that the frequency of the jumps increased very rapidly with the stress as well as with the temperature. It appears that the temperature dependence is of the same character as that of a chemical reaction. This behaviour is in striking agreement with a theoretical treatment given by Prof. R. Becker⁽²⁾.

Let σ_m be the microscopic shear strength of the glide plane, σ the average stress and q the stress concentration factor at the point where gliding is to start; thus $q\sigma$ is the local stress there. A glide process will start before $q\sigma$ reaches the value σ_m , if a thermal fluctuation of stress raises $q\sigma$ to the value of σ_m in a sufficiently large volume. If A is the work necessary for this, then the probability of such a critical fluctuation will be, according to the theorem of Boltzmann, $p = Ce^{-A/kT}$. The work necessary for raising the microscopic shear stress $q\sigma$ to the value σ_m of the microscopic shear strength in a volume V is $A = (\sigma_m - q\sigma)^2 V / 2G$, where G is the modulus of shear; thus the probability of a critical fluctuation is

$$p = C \exp \left[- \frac{(\sigma_m - q\sigma)^2 V}{2GkT} \right]. \quad \dots(8)$$

This formula gives a very steep increase of p with the stress and the temperature, as is shown by the jerky extension of zinc.

A somewhat puzzling feature of the Becker formula (8) is the volume V . The physical interpretation of this quantity is as follows. Suppose a glide process to start at the point P in figure 3, and the radius of the dislocation line to be r . The glide process is connected with a decrease of elastic energy, since the parts of the crystal adjoining the dislocated area have become free from stress; the decrease is proportional to the area and to the depth of the unstressed zone, which may be taken as proportional to r . Thus the change of elastic energy is $E_e = -Kr^3$. On the other hand, the dislocation zone represents an energy, consisting of an elastic part and of the mutual energy of atoms which are out of phase with respect to the opposite lattice plane. This energy E_d is proportional to the length of the dislocation line: $E_d = Lr$. The glide process will be stable if the change of energy for an infinitesimal change of r vanishes:

$$d(E_e + E_d)/dr = -3Kr^2 + L = 0, \quad \dots(6)$$

or

$$r = \sqrt{(L/3K)}.$$

The stress fluctuation must extend over an area r^2 of the glide plane, that is, over a volume $V \sim r^3$, in order to produce a stable dislocation; otherwise it is probable that the dislocation will not spread but vanish.

§ 10. THE RATE OF FLOW

The rate of glide flow will be determined, according to (4), by the number N of dislocations per unit volume and by their rate of propagation. The dislocations may be produced by critical stress fluctuations at stress concentrations; they may be present in the crystal from previous deformation or formed during the growth of the crystal.

As regards the propagation of dislocations, it is obvious that this process is connected with jumps across potential barriers, although these are much lower than the potential barriers of homogeneous gliding. Let σ_a be the average stress necessary to set a dislocation into movement in the absence of thermal movement. If the average stress σ is less than σ_a , the dislocation can still move with the help of thermal fluctuations; its velocity will be given by a formula of the type of (4), for example:

$$v = C \exp \left[-\frac{(\sigma_a - \sigma)^2 V}{2GkT} \right]. \quad \dots\dots(10)$$

Here, however, we meet a problem which did not arise in the case of glide processes starting from stress concentrations. In a stress concentration, the shear stress $q\sigma$ certainly cannot exceed the microscopic shear strength σ_m ; its highest possible value is σ_m , in which case gliding will start instantaneously, without having to wait for the help of a thermal fluctuation. A dislocation, however, will not run out of the crystal with infinite velocity if an average stress exceeding σ_a is applied; we must consider, therefore, the case when the average stress is higher

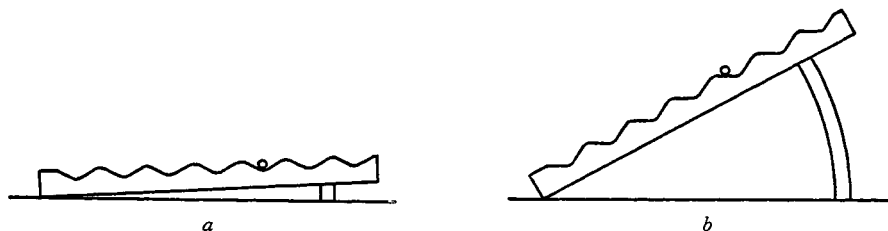


Figure 8.

than the critical stress necessary for moving the dislocation without thermal help. In this case the mechanism of propagation will be quite different from that described by (10); the two mechanisms and the transition between them can be illustrated by a simple model.

Let the movement of a dislocation along a periodical potential field be schematically represented by the movement of a ball along a board with a wavy surface (figure 8), the ball being prevented by a guide from leaving the surface of the board. The application of a moderate stress is represented by a slight inclination of the board (figure 8a). The ball cannot leave a trough unless a thermal fluctuation throws it into the next. With the energy acquired the ball would be able to continue its movement without further thermal help if the board were rigid. Otherwise a part of its energy would be dissipated in the form of elastic waves and its movement would soon be arrested. Its mean velocity is proportional to the frequency of critical thermal fluctuations and is given, therefore, by a Boltzmann formula, with the value of the energy barrier between two troughs as A .

Now let the board be tilted until the potential barriers disappear (figure 8b). With increasing velocity the pulsating force exerted by the ball upon the board increases, and with it the energy of the elastic waves imparted at every step to the board. Finally the energy dissipated per step becomes equal to the potential difference

between two steps and then a stationary state is reached with a constant mean velocity which depends on the potential difference between two steps.

From the behaviour of this model we recognize that dislocations will have a well-defined velocity less than the velocity of sound even if the stress applied is higher than the critical stress σ_d necessary for moving dislocations without thermal help. This velocity is determined by a mechanism of momentum transfer between the gliding parts of the crystal; as the stress increases, the mechanism of the propagation of gliding will change over from the thermal activation mechanism expressed by (10) to this momentum transfer mechanism.

§ 11. VARIATION OF THE YIELD STRESS WITH TEMPERATURE

In the neighbourhood of the yield point the mechanism of the propagation of dislocations is probably a pure thermal activation mechanism. Whether the rate of plastic flow is mainly determined by the number of glide processes starting per unit time, or by the velocity of the dislocations, we may expect it to show the dependence upon stress and temperature expressed by the formulae (4) or (10).*

That the rate of flow increases very rapidly with stress is well known. That the equally rapid increase with temperature required by (4) or (10) actually exists was felt to be rather surprising at the time when the theory was put forward. In fact, the extremely steep temperature dependence following from (4) or (10) brought these formulae into discredit for many years; they were considered as refuted by the fact that metal crystals are plastic even at the lowest temperatures. This conclusion, however, was erroneous. By a relatively small increase of σ we can maintain the value of $(\sigma_m - q\sigma)^2/T$ and thus make the rate of flow invariant for any decrease of temperature.

From (4) or (10) a formula giving the variation of the yield stress with the temperature can be derived⁽¹²⁾. The yield stress is the stress at which the rate of flow reaches the limit Y beyond which it becomes easily observable. Thus using, for example, (10), the condition of yielding is

$$Y = \text{const.} = C \exp \left[-\frac{(\sigma_d - \sigma)^2 V}{2GkT} \right]. \quad \text{.....(11)}$$

Assuming that V and G are approximately constant, we have

$$(\sigma_d - \sigma)^2/T = \text{const.} \quad \text{.....(11a)}$$

or

$$\sigma = \sigma_d - B\sqrt{T}. \quad \text{.....(12)}$$

The experimental results agree very well with this formula which, at present, is the only theoretical result relating to the dynamics of gliding that can be tested quantitatively by the measurements available.

* For very small stresses (4) or (10) obviously cannot give the rate of flow; it is easy, however, to give a generalized formula which applies in this case also⁽¹⁶⁾.

§ 12. THEORETICAL SIGNIFICANCE OF THE YIELD STRESS

The assumption made in deriving (12), that V is nearly constant, requires some explanation. If the process that determines the rate of flow is the creation of dislocations at stress concentrations, then K in (9), and thus V , will depend on the stress. If, on the other hand, the rate of flow is mainly determined by the velocity of dislocations, then the linear dimensions of V may be determined by the thickness of the dislocation zone and they may be more or less independent of σ . The success of formula (12) indicates that the yield point is the stress at which the dislocations begin to be mobile, not the stress at which the number of glide processes started at stress concentrations begin to increase rapidly.

This conclusion is supported by other observations. First, the experiments mentioned show that a zinc crystal, hardened by the removal of effective dislocations and flaws, regains its usual mode of yielding and its usual yield point after plastic deformation has taken place, i.e. after dislocations have been produced. Second, it is well known that the yield point cannot be influenced like the tensile strength of brittle materials by altering the state of the surface. All this points to the conclusion that the yield stress is the stress at which the dislocations become mobile, or capable of producing avalanches of new dislocations, in the sense of § 8.

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